

## Pinecone setup + test

Go to Pinecone website => start for free

<https://www.pinecone.io/>

Create API key, Index

To test in local and manually

Run

generate\_logs.py

create\_evidence\_chunk.py

ingest\_to\_pinecone.py

```
Submitted 2 records to Pinecone
UpsertRecordsResponse(record_count=2, response_info=ResponseInfo(raw_headers={'date': 'Wed
, 24 Jun 2026 01:58:57 GMT', 'content-length': '0', 'connection': 'keep-alive', 'x-pinecon
e-request-lsn': '1', 'x-pinecone-api-version': '2025-10', 'x-envoy-upstream-service-time':
'616', 'x-pinecone-response-duration-ms': '617', 'server': 'envoy'})))
(rag-lc) PS C:\Users\saito\source\repos\AIEngineering_JavaPython>
```

BROWSERMETRICSNAMESPACES (1)IMPORTSCONFIGURATION

Records

Upsert record

Namespace

Operation

Text

simulated-prod

Search by text

Hello, world!

Top

K

10

+ Filter

+ Rerank

Search

Search: 2 results (top\_k=10)

7 embed tokens 1 RUs

1

SCORE

0.0587

\_id: trace-TRACE-10045

environment: "simulated-prod"

errorCodes: "FRAUD,TIMEOUT,POD\_RESTARTED,SERVICE\_UNHEALTHY"

services: "account-service,auth-service,fraud-service,payment-service"

text: "Evidence logs for traceId TRACE-10045 and transactionId TXN-789.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-service | level=INFO | ev...

traceId: "TRACE-10045"

transactionId: "TXN-789"

2

SCORE

0.0570

\_id: trace-TRACE-10046

environment: "simulated-prod"

errorCodes: "BALANCE\_CHECK\_FAILED,DB\_CONNECTION\_ERROR,DB\_POOL\_EXHAUSTED"

services: "account-service,auth-service,notification-service,payment-service"

text: "Evidence logs for traceId TRACE-10046 and transactionId TXN-790.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO | ev...

traceId: "TRACE-10046"

transactionId: "TXN-790"

(rag-1c) PS C:\Users\saito\source\repos\AIEngineering\_JavaPython\AI-Banking-Logs-RCA-Agent\search> python .\test\_search.py

Question:

Why are payments failing?

Pinecone Search Results:

-----  
Score: None

ID: None

Trace ID: TRACE-10046

Transaction ID: TXN-790

Services: account-service,auth-service,notification-service,payment-service

Error Codes:

BALANCE\_CHECK\_FAILED,DB\_CONNECTION\_ERROR,DB\_POOL\_EXHAUSTED

Environment: simulated-prod

Evidence Text:

Evidence logs for traceId TRACE-10046 and transactionId TXN-790.

The following logs show the request flow in timestamp order:

2026-06-22T10:17:29Z | service=auth-service | level=INFO | eventType=APPLICATION\_LOG |  
errorCode=NONE | latencyMs=105 | message=Token validation successful

2026-06-22T10:17:30Z | service=account-service | level=ERROR |  
eventType=APPLICATION\_ERROR | errorCode=DB\_CONNECTION\_ERROR | latencyMs=5100 |  
message=Account balance lookup failed because database connection timed out

2026-06-22T10:17:31Z | service=payment-service | level=ERROR |  
eventType=APPLICATION\_ERROR | errorCode=BALANCE\_CHECK\_FAILED | latencyMs=5200 |  
message=Payment failed because account balance could not be verified

2026-06-22T10:17:32Z | service=account-service | level=ERROR | eventType=DATABASE\_ERROR |  
errorCode=DB\_POOL\_EXHAUSTED | latencyMs=5000 | message=Connection pool exhausted while  
connecting to account database

2026-06-22T10:17:33Z | service=notification-service | level=WARN |  
eventType=APPLICATION\_LOG | errorCode=NONE | latencyMs=80 | message=Payment failure  
notification queued for customer

-----  
Score: None

ID: None

Trace ID: TRACE-10045

Transaction ID: TXN-789

Services: account-service,auth-service,fraud-service,payment-service

Error Codes: FRAUD\_TIMEOUT,POD\_RESTARTED,SERVICE\_UNHEALTHY

Environment: simulated-prod

Evidence Text:

Evidence logs for traceId TRACE-10045 and transactionId TXN-789.

The following logs show the request flow in timestamp order:

2026-06-22T10:15:29Z | service=auth-service | level=INFO | eventType=APPLICATION\_LOG |  
errorCode=NONE | latencyMs=120 | message=Token validation successful  
2026-06-22T10:15:30Z | service=account-service | level=INFO | eventType=APPLICATION\_LOG |  
errorCode=NONE | latencyMs=210 | message=Account balance check successful  
2026-06-22T10:15:31Z | service=payment-service | level=ERROR |  
eventType=APPLICATION\_ERROR | errorCode=FRAUD\_TIMEOUT | latencyMs=4300 |  
message=Payment failed due to fraud-service timeout  
2026-06-22T10:15:33Z | service=fraud-service | level=ERROR |  
eventType=READINESS\_PROBE\_FAILED | errorCode=SERVICE\_UNHEALTHY | latencyMs=4200  
| message=Readiness probe failed: /health returned 503  
2026-06-22T10:15:34Z | service=fraud-service | level=WARN | eventType=POD\_RESTARTED |  
errorCode=POD\_RESTARTED | latencyMs=0 | message=fraud-service pod restarted after repeated  
health check failures

Records

Upsert n

Namespace

Operation

Text

simulated-prod

Search by text

why are payments failing?

+ Filter

+ Rerank

Search

Search: 2 results (top\_k=10)

8 embed tokens

1

SCORE  
0.2098

\_id: trace-TRACE-10046

environment: "simulated-prod"

errorCodes: "BALANCE\_CHECK\_FAILED,DB\_CONNECTION\_ERROR,DB\_POOL\_EXHAUSTED"

services: "account-service,auth-service,notification-service,payment-service"

text: "Evidence logs for traceld TRACE-10046 and transactionId TXN-790. The following logs show the request flow in timestamp order: 2026-06-22T10:17:29Z | service=auth-service | level=INFO | ev...

traceld: "TRACE-10046"

transactionId: "TXN-790"

2

SCORE  
0.1527

\_id: trace-TRACE-10045

environment: "simulated-prod"

errorCodes: "FRAUD\_TIMEOUT,POD\_RESTARTED,SERVICE\_UNHEALTHY"

services: "account-service,auth-service,fraud-service,payment-service"

text: "Evidence logs for traceld TRACE-10045 and transactionId TXN-789. The following logs show the request flow in timestamp order: 2026-06-22T10:15:29Z | service=auth-service | level=INFO | ev...

traceld: "TRACE-10045"

transactionId: "TXN-789"

Namespace

simulated-prod

Operation

Search by text

Text

Which trace shows fraud-service becoming unhealthy?

Top

K

+ Filter

+ Rerank

Search

Search: 2 results (top\_k=10)11 embed tokens 11

1

SCORE

0.3355

\_id: trace-TRACE-10045

environment: "simulated-prod"

errorCodes: "FRAUD\_TIMEOUT,POD\_RESTARTED,SERVICE\_UNHEALTHY"

services: "account-service,auth-service,fraud-service,payment-service"

text: "Evidence logs for traceld TRACE-10045 and transactionId TXN-789.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:15:29Z | service=auth-service | level=INFO | ev...

traceld: "TRACE-10045"

transactionId: "TXN-789"

2

SCORE

0.2338

\_id: trace-TRACE-10046

environment: "simulated-prod"

errorCodes: "BALANCE\_CHECK\_FAILED,DB\_CONNECTION\_ERROR,DB\_POOL\_EXHAUSTED"

services: "account-service,auth-service,notification-service,payment-service"

text: "Evidence logs for traceld TRACE-10046 and transactionId TXN-790.\n\nThe following logs show the request flow in timestamp order:\n\n2026-06-22T10:17:29Z | service=auth-service | level=INFO | ev...

traceld: "TRACE-10046"

transactionId: "TXN-790"

## Automating with AWS

Ran python cloud/push\_logs\_to\_cloudwatch.py

Go to AWS CloudWatch



### Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q Filter events - press enter to search

|   | Timestamp                | Message                                                                                |
|---|--------------------------|----------------------------------------------------------------------------------------|
|   |                          | No older events at this moment. <a href="#">Retry</a>                                  |
| ▶ | 2026-06-22T10:15:29.000Z | { "timestamp": "2026-06-22T10:15:29Z", "service": "auth-service", "level": "INFO", "tr |
| ▶ | 2026-06-22T10:15:30.000Z | { "timestamp": "2026-06-22T10:15:30Z", "service": "account-service", "level": "INFO",  |
| ▶ | 2026-06-22T10:15:31.000Z | { "timestamp": "2026-06-22T10:15:31Z", "service": "payment-service", "level": "ERROR", |
| ▶ | 2026-06-22T10:15:33.000Z | { "timestamp": "2026-06-22T10:15:33Z", "service": "fraud-service", "level": "ERROR", " |
| ▶ | 2026-06-22T10:15:34.000Z | { "timestamp": "2026-06-22T10:15:34Z", "service": "fraud-service", "level": "WARN", "t |
| ▶ | 2026-06-22T10:17:29.000Z | { "timestamp": "2026-06-22T10:17:29Z", "service": "auth-service", "level": "INFO", "tr |
| ▶ | 2026-06-22T10:17:30.000Z | { "timestamp": "2026-06-22T10:17:30Z", "service": "account-service", "level": "ERROR", |
| ▶ | 2026-06-22T10:17:31.000Z | { "timestamp": "2026-06-22T10:17:31Z", "service": "payment-service", "level": "ERROR", |
| ▶ | 2026-06-22T10:17:32.000Z | { "timestamp": "2026-06-22T10:17:32Z", "service": "account-service", "level": "ERROR", |
| ▶ | 2026-06-22T10:17:33.000Z | { "timestamp": "2026-06-22T10:17:33Z", "service": "notification-service", "level": "WA |
|   |                          | No newer events at this moment. <i>Auto retry paused.</i> <a href="#">Resume</a>       |

### Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn mo](#)

Q Filter events - press enter to search

|   | Timestamp                | Message                                                                                                                                                                                                                                                                                                                                                                                 |
|---|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                          | No older events at this moment. <a href="#">Retry</a>                                                                                                                                                                                                                                                                                                                                   |
| ▼ | 2026-06-22T10:15:29.000Z | <pre>{ "timestamp": "2026-06-22T10:15:29Z", "service": "auth-ser  {   "timestamp": "2026-06-22T10:15:29Z",   "service": "auth-service",   "level": "INFO",   "traceId": "TRACE-10045",   "transactionId": "TXN-789",   "eventType": "APPLICATION_LOG",   "message": "Token validation successful",   "errorCode": "NONE",   "latencyMs": 120,   "environment": "simulated-prod" }</pre> |
| ▶ | 2026-06-22T10:15:30.000Z | { "timestamp": "2026-06-22T10:15:30Z", "service": "account-                                                                                                                                                                                                                                                                                                                             |

## Next Create Lambda function

☰ [Lambda](#) > [Functions](#) > Create function

### Create function [Info](#)

Choose one of the following options to create your function.

☒ **Author from scratch**  
Start with a simple Hello World example.

☐ **Use a blueprint**  
Build a Lambda application from sample code and configuration presets for common use cases.

☐ **Container image**  
Select a container image to deploy f

---

#### Basic information

**Function name**  
Enter a name that describes the purpose of your function.

banking-log-ingestion-lambda

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (\_).

**Runtime** | [Info](#)  
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.14

**Permissions**  
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution** role under **Additional settings**.

---

#### Custom settings [Info](#)

**Durable execution - [new](#)** 

Build resilient, stateful applications with automatic failure recovery.

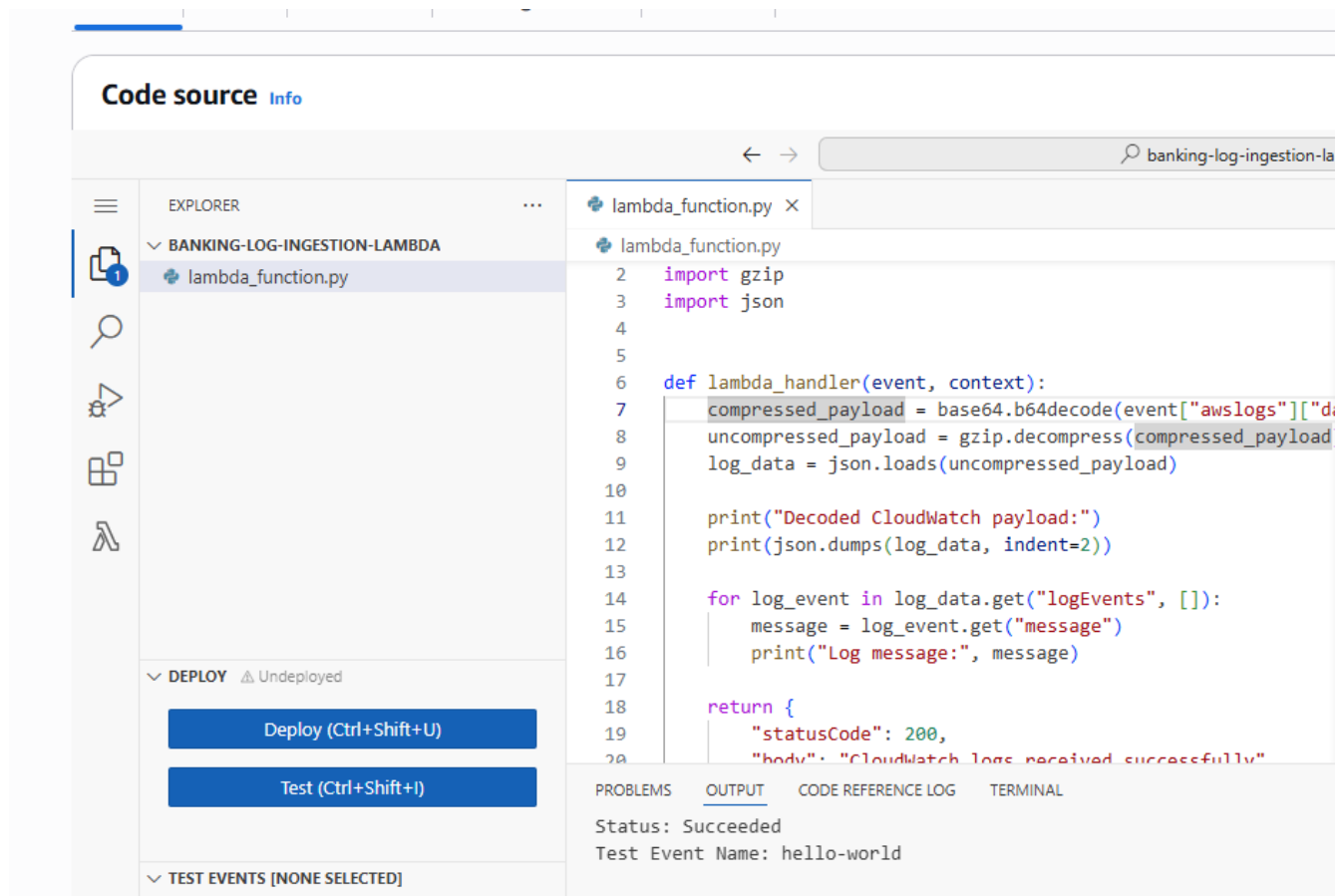
☐

**EC2 capacity provider - [new](#)** 

Run Lambda on your preferred EC2 instance types instead of Lambda's serve

Copy paste the lambda\_function.py

Click Deploy



After deploying, go back to CloudWatch

Go to Subscription filters of /banking/simulated-prod



/banking/simulated-prod

ActionsView in Logs InsightsStart tailingSearch log group

▼ Log group details

Log class  
Standard

ARN  
arn:aws:logs:ca-central-1:577638386543:log-group:/banking/simulated-prod:\*

Creation time  
34 minutes ago

Retention  
Never expire

Stored bytes  
-

Metric filters  
0

Subscription filters  
0

Contributor Insights rules  
-

KMS key ID  
-

Deletion protection  
Off

Data protection  
-

Sensitive data count  
-

Custom field indexes  
Configure

Transformer  
Configure

Anomaly detection  
Configure

Bearer token authentication  
Off

Syslog ingestion  
InfoConfigure

Log streamsTagsData protectionAnomaly detectionMetric filtersSubscription filtersContributor InsightsField indexesTransformer

Subscription filters (0) Info

Account-level and log-group-level subscription filters are additive. A log group can have up to 1 account-level and up to 2 log-group-level subscription filters.

Filter name

Filter pattern

Destination ARN

Subscription filter type

Applied on transformed ...

Field selection criteria

Emit system fields

There are no subscription filters.

# Create Lambda Subscription filter

## Create Lambda subscription filter

You are about to start streaming data from your `/banking/simulated-prod` log group to an Amazon Lambda function. Any new log data sent to this log group will be sent to the function you choose.

### Choose destination

Choose the Lambda function to execute when a log event matches the filter you are going to specify. [Learn more about Lambda functions.](#)

#### Lambda function

Select the Lambda function you want to subscribe to the filter.

 Streaming large amounts of CloudWatch Logs data to other destinations might result in high usage charges. We recommend that you create a Budget in the Billing and Cost Management console. For more information, see: [Managing Your Costs with Budgets](#)

### Configure log format and filters

Choose your log format to get a recommended filter pattern for your log data, or select "Other" to enter a custom filter pattern. An empty filter pattern matches all log events.

#### Log format

#### Subscription filter pattern

Specify the log event structure and any filter conditions to apply on your log data as it gets streamed to the Amazon Lambda service.

#### Subscription filter name

A subscription filter is the automatic bridge between CloudWatch Logs and Lambda.

Subscription filter name: banking-logs-to-lambda

Click Start streaming

New log arrives in `/banking/simulated-prod`

↓

Subscription filter checks it

↓

If it matches the filter pattern

↓

CloudWatch sends it to Lambda

↓

Lambda runs automatically

Re-run `push_logs_to_cloudwatch.py`

Go to `banking-log-ingestion-lambda`

```
< /aws/lambda/banking-log-ingestion-lambda
```

### Sensitive data count

| <input type="checkbox"/> | Log stream                                                            | Last event time  |
|--------------------------|-----------------------------------------------------------------------|------------------|
| <input type="checkbox"/> | <a href="#">2026/06/24/[\$LATEST]23d22156b2cf4bc6b3df0e0d81a30175</a> | 2026-06-24 14:34 |
| <input type="checkbox"/> | <a href="#">2026/06/24/[\$LATEST]d59b245ff70647a7ad19547ca8743feb</a> | 2026-06-24 14:34 |

|   |                          |                                                                                                                                                                                                                                                                                                                                              |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ▶ | 2026-06-24T14:42:11.619Z | "timestamp": 1782123453000,                                                                                                                                                                                                                                                                                                                  |
| ▶ | 2026-06-24T14:42:11.619Z | "message": "{\"timestamp\": \"2026-06-22T10:17:33Z\", \"service\": \"notification-                                                                                                                                                                                                                                                           |
| ▶ | 2026-06-24T14:42:11.619Z | }                                                                                                                                                                                                                                                                                                                                            |
| ▶ | 2026-06-24T14:42:11.619Z | ]                                                                                                                                                                                                                                                                                                                                            |
| ▶ | 2026-06-24T14:42:11.619Z | }                                                                                                                                                                                                                                                                                                                                            |
| ▼ | 2026-06-24T14:42:11.619Z | Log message: {"timestamp": "2026-06-22T10:15:29Z", "service": "auth-service", "lev                                                                                                                                                                                                                                                           |
|   |                          | Log message:<br><pre> {   "timestamp": "2026-06-22T10:15:29Z",   "service": "auth-service",   "level": "INFO",   "traceId": "TRACE-10045",   "transactionId": "TXN-789",   "eventType": "APPLICATION_LOG",   "message": "Token validation successful",   "errorCode": "NONE",   "latencyMs": 120,   "environment": "simulated-prod" } </pre> |
| ▶ | 2026-06-24T14:42:11.619Z | Log message: {"timestamp": "2026-06-22T10:15:30Z", "service": "account-service", "                                                                                                                                                                                                                                                           |
| ▶ | 2026-06-24T14:42:11.619Z | Log message: {"timestamp": "2026-06-22T10:15:31Z", "service": "payment-service", "                                                                                                                                                                                                                                                           |

Local script → CloudWatch Logs → Subscription Filter → Lambda 

Add Pinecone environment variables to Lambda

Lambda → banking-log-ingestion-lambda → Configuration → Environment variables

Layers



CloudWatch Logs

+ Add trigger

Code

Test

Monitor

Configuration

Aliases

Versions

## Configuration

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

### General configuration [Info](#)

Description

-

Timeout

0 min 3 sec

## Edit environment variables

### Environment variables

You can define environment variables as key-value pairs that are accessible from your function code. These are useful to store configuration information.

Key

PINECONE\_INDEX\_NAME

Value

banking-logs-rca

PINECONE\_NAMESPACE

simulated-prod

[Add environment variable](#)

► **Encryption configuration**

Go to vscode

```
mkdir lambda_build
```

```
mkdir lambda_build\package
```

```
python -m pip install pinecone -t lambda_build\package
```

Add test lambda\_function.py

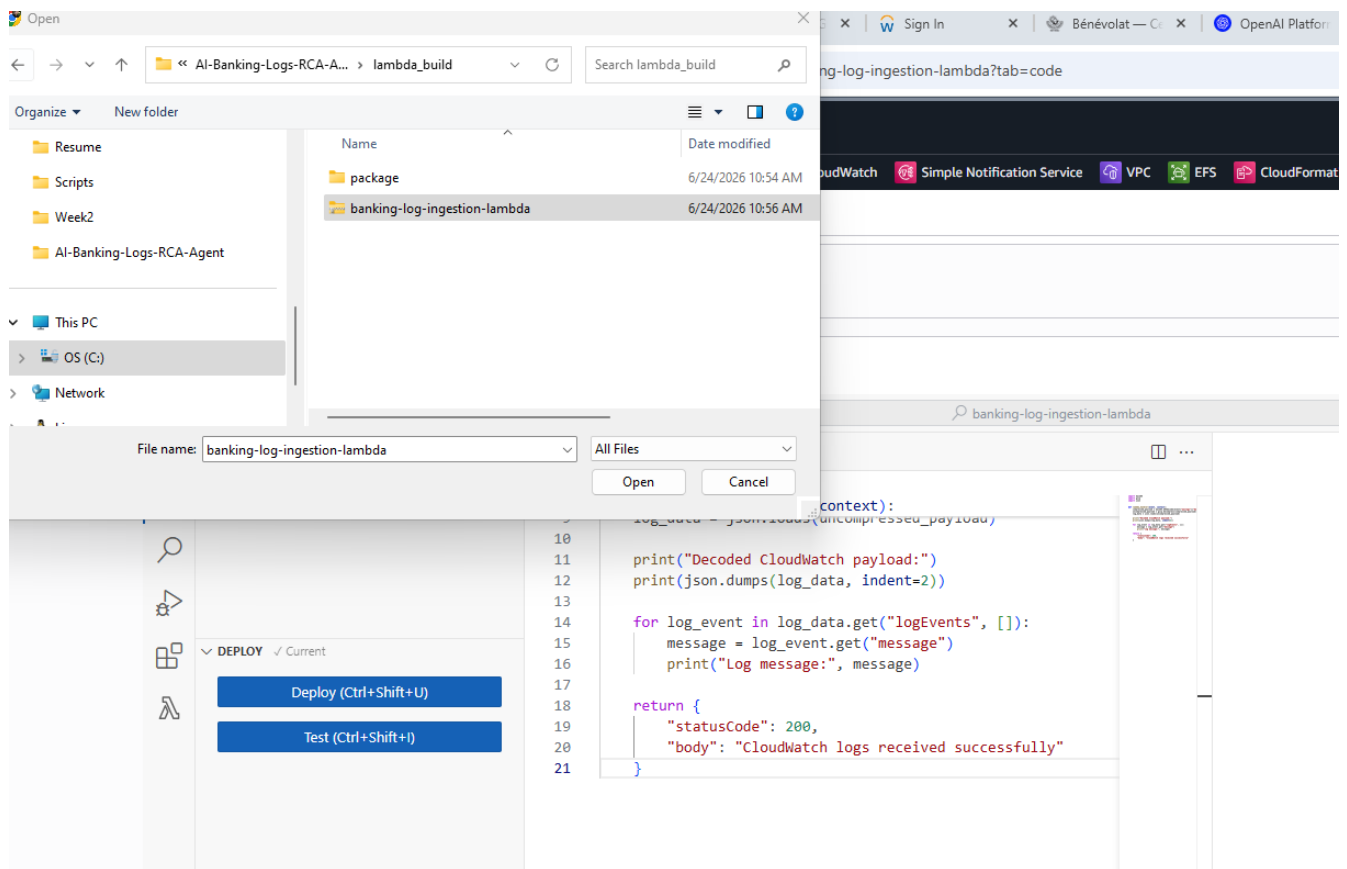
```
cd lambda_build\package
```

```
Compress-Archive -Path * -DestinationPath ..\banking-log-ingestion-lambda.zip -Force
```

```
cd ..\..
```

```
python cloud/push_logs_to_cloudwatch.py
```

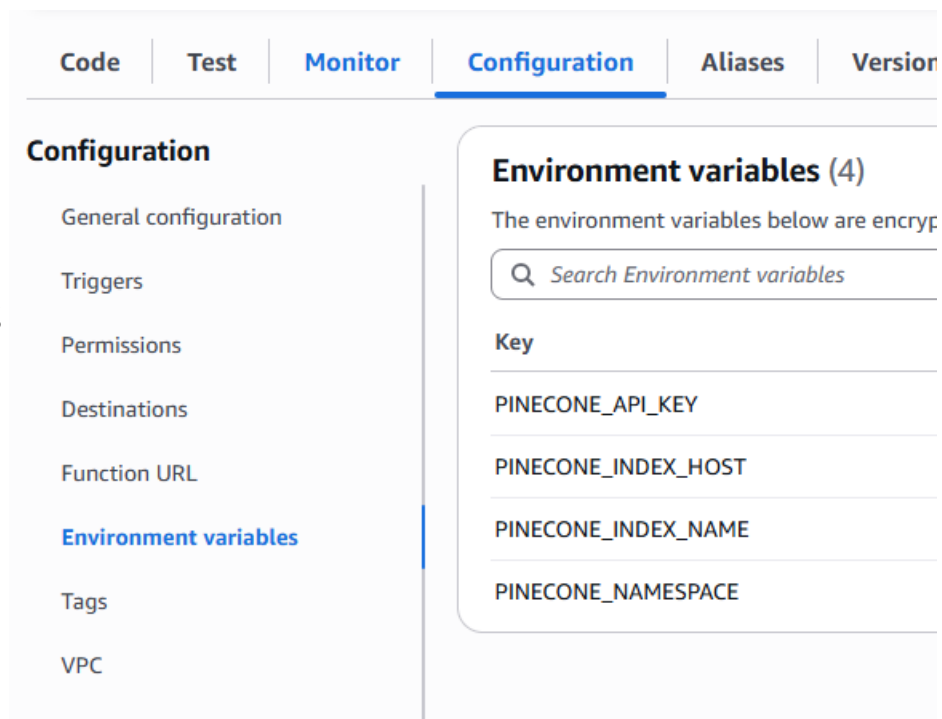
I upload the new zip file into Lambda



Now I retry with `deploy_lambda_zip.py`

`python cloud/deploy_lambda_zip.py`

Make sure you have following Environment variables in Lambda



Re-run push\_logs\_to\_cloudwatch.py

management > /aws/lambda/banking-log-ingestion-lambda > 2026/06/24/[\$LATEST]f6ae2a344c0e42479af8c6475e146183

<

▶

### Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Clear

1m

30m

1h

| ▶                                                                         | Timestamp                | Message                                                                                                                         |
|---------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| No older events at this moment. <a href="#">Retry</a>                     |                          |                                                                                                                                 |
| ▶                                                                         | 2026-06-24T16:09:17.695Z | INIT_START Runtime Version: python:3.12.mainlinev2.v11 Runtime Version ARN: arn:aws:lambda:ca-central-1::runtime:0eb6fb00a97c1a |
| ▶                                                                         | 2026-06-24T16:09:17.831Z | START RequestId: 6b30cb3b-edd3-4c86-a331-1b6dfe726c2d Version: \$LATEST                                                         |
| ▶                                                                         | 2026-06-24T16:09:17.831Z | REST PINECONE INGESTION VERSION IS RUNNING                                                                                      |
| ▶                                                                         | 2026-06-24T16:09:17.832Z | Decoded CloudWatch payload successfully                                                                                         |
| ▶                                                                         | 2026-06-24T16:09:17.832Z | Parsed 10 JSON log messages                                                                                                     |
| ▶                                                                         | 2026-06-24T16:09:17.832Z | Created 2 evidence records                                                                                                      |
| ▶                                                                         | 2026-06-24T16:09:18.653Z | Pinecone response status: 201                                                                                                   |
| ▶                                                                         | 2026-06-24T16:09:18.653Z | Pinecone response body:                                                                                                         |
| ▶                                                                         | 2026-06-24T16:09:18.655Z | END RequestId: 6b30cb3b-edd3-4c86-a331-1b6dfe726c2d                                                                             |
| ▶                                                                         | 2026-06-24T16:09:18.655Z | REPORT RequestId: 6b30cb3b-edd3-4c86-a331-1b6dfe726c2d Duration: 823.70 ms Billed Duration: 956 ms Memory Size: 128 MB Max Mem  |
| No newer events at this moment. Auto retry paused. <a href="#">Resume</a> |                          |                                                                                                                                 |

- Local raw\_logs.jsonl
- CloudWatch Logs
  - Subscription Filter
  - Lambda
  - Pinecone REST upsert
- ✓



